

The Planck Depth and Optimal Trigger Design: Composition-Inflation Applied to Real-Time Event Selection at the Large Hadron Collider

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Abstract

The Large Hadron Collider produces proton–proton collisions at 40.079 MHz. The trigger system must reduce this rate to approximately 1 kHz for permanent storage — a rejection fraction of 99.997 % in under 2.5 μ s. We demonstrate that this design problem has a mathematically natural solution grounded in the composition-inflation formula $T(n, d) = d(d + 1)^{n-1}$ derived from the partition algebra of bounded phase space.

The central result is the *Planck depth* n_P for the LHC bunch clock: the minimum number of bunch crossings after which the categorical trajectory count $T(n_P, d)$ equals or exceeds the ratio τ_B/t_P of the bunch-crossing period to the Planck time. For the LHC at $d = 3$ independent detector subsystems and $\nu_B = 40.079$ MHz, we derive $n_P = 60$ exactly. The current L1 trigger latency of 100 bunch crossings operates at $n/n_P = 1.67$, placing it in the super-Planck categorical resolution regime.

We derive three additional results with direct engineering implications. First, the 25 ns bunch-crossing window is not an arbitrary engineering choice: it equals the partition propagation time $\tau_c = \Delta x/c$ for the ATLAS detector radius, and is therefore determined by the geometry of the interaction region and the speed of light, both of which are consequences of the partition framework. Second, the partition uncertainty relation $\Delta\mathcal{M} \cdot \tau_p \geq \hbar$ implies a minimum detectable partition depth change of 26 eV per bunch crossing, which is the fundamental resolution floor of the trigger — not an engineering parameter. Third, the HL-LHC upgrade adding a fourth independent detector subsystem ($d : 3 \rightarrow 4$) reduces the Planck depth from 60 to 52, recovering eight bunch crossings (200 ns) of categorical resolution headroom — a quantitative prediction derivable from first principles without any knowledge of the specific detector technology.

We also derive the *composition efficiency* η_C and the *Planck Utilisation Index* (PUI) as dimensionless metrics characterising how much of the available categorical resolution the trigger exploits. The current ATLAS trigger menu of approximately 10^3 paths achieves $\eta_C \approx 10\%$:

90% of the theoretically available categorical resolution remains unused. These metrics provide a framework-grounded basis for trigger optimisation that does not depend on Monte Carlo simulation of physics processes.

Keywords: LHC trigger; composition-inflation; Planck depth; bunch crossing; categorical resolution; trigger design; HL-LHC; partition algebra; timing window; detector subsystems

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1 Introduction

1.1 The event selection problem

The Large Hadron Collider at CERN delivers proton-proton collisions at a rate of 40.079 MHz (Evans and Bryant, 2008). Each collision event, if fully recorded, produces approximately 1–2 MB of raw detector data across the ATLAS and CMS detectors (ATLAS Collaboration, 2008; CMS Collaboration, 2008). The aggregate data rate is thus of order 80 TB per second — several orders of magnitude beyond any available storage technology.

The trigger system must reduce this rate in real time. The current ATLAS Level-1 (L1) trigger accepts approximately 100 kHz, which the High-Level Trigger (HLT) then reduces to approximately 1 kHz for permanent storage (ATLAS Collaboration, 2020b). The L1 decision must be made within a fixed pipeline latency of $2.5 \mu\text{s}$ = 100 bunch-crossing periods, after which the detector front-end buffers are overwritten.

The event selection problem therefore has three constraints that are simultaneously tight, independent, and non-negotiable:

- (i) *Rate*: accept ≤ 100 kHz from 40.079 MHz input ($\leq 0.25\%$).
- (ii) *Latency*: decide within 100 bunch crossings ($2.5 \mu\text{s}$).
- (iii) *Irreversibility*: discarded events are permanently lost.

Existing formal frameworks for trigger design are empirical and simulation-based: the trigger menu is a list of physics-motivated conditions, each calibrated against Monte Carlo simulations of signal and background processes. The theoretical question — *what is the maximum number of distinct event categories a trigger of given depth can discriminate?* — has not previously been answered from first principles. This paper answers it.

1.2 Summary of results

The key results, all derived from the partition algebra of bounded phase space without additional physical postulates:

- (1) **Trigger State Inflation** (Theorem 2.2): a trigger with d independent subsystems and n bunch crossings can discriminate $T(n, d) = d(d+1)^{n-1}$ distinct event categories.
- (2) **LHC Planck Depth** (Theorem 3.3): $n_P = 60$ for the LHC at $d = 3$.
- (3) **Timing Window Derivation** (Theorem 4.1): the 25 ns bunch-crossing window equals the partition propagation time across the ATLAS detector interaction region.
- (4) **Partition Uncertainty** (Theorem 4.3): $\Delta\mathcal{M} \cdot \tau_p \geq \hbar$ implies a 26 eV minimum detectable energy difference per bunch crossing.
- (5) **HL-LHC Dimension Advantage** (Theorem 8.1): $d : 3 \rightarrow 4$ reduces $n_P : 60 \rightarrow 52$, saving 8 crossings.
- (6) **Categorical Efficiency Metrics** (Definitions 9.1–9.4): current ATLAS trigger operates at $\eta_C \approx 10\%$, PUI = 1.67.
- (7) **QND Multi-Subsystem Classification** (Theorem 5.4): all four detector subsystem observables commute, enabling simultaneous classification without quantum backaction.
- (8) **Single-Particle Thermodynamics** (Theorem 6.2): each detected particle satisfies $PV = k_B T_{\text{cat}}$ where $T_{\text{cat}} = \hbar\omega_c/(2\pi k_B)$.

1.3 Organisation

Section 2 derives the trigger state inflation formula. Section 3 derives the Planck depth and validates it against the LHC parameters. Section 4 establishes the physical origin of the timing window. Section 5 maps detector subsystems to commuting partition coordinates. Section 6 develops single-particle thermodynamics for the trigger. Section 7 develops trigger design principles from the theory. Section 8 analyses the HL-LHC upgrade. Section 9 defines the efficiency metrics. Section 10 discusses implications and open questions.

All theorems are proved from first principles or cited from the companion mathematical foundation paper (Sachikonye, d). The paper is self-contained: every symbol is defined at first use.

2 Composition-Inflation Applied to Event Selection

2.1 The naive state count and why it fails

A trigger system with d independent detector subsystems observing one bunch crossing assigns each crossing to one of at most d single-subsystem categories. Over n crossings, the naive counting — independent binary decisions — gives 2^n possible outcomes.

This count is wrong. It treats each crossing as independent and ignores the combinatorial structure of how n crossings can be distributed among categories. The correct count is derived from the number of *labeled integer compositions* of n in d dimensions, which is qualitatively different (exponential with base $d + 1$, not 2).

2.2 Labeled compositions and the trigger

Definition 2.1 (Trigger Labeled Composition). A *trigger labeled composition* of n bunch crossings in d detector dimensions is a pair $(\mathbf{C}, \mathbf{L})$ where $\mathbf{C} = (c_1, \dots, c_k)$ is an ordered partition of n into k positive integer parts and $\mathbf{L} = (\ell_1, \dots, \ell_k)$ with $\ell_i \in \{1, \dots, d\}$ specifies which detector subsystem characterises the i -th group of crossings.

The physical meaning is as follows. A trigger that records which detector subsystem dominates each contiguous group of crossings — rather than treating each crossing independently — captures the temporal structure of the event. An event that fires the EM calorimeter for 3 crossings then the muon system for 2 crossings is categorically distinct from one that fires the muon system first, even if the total crossing counts are the same. The composition captures this ordering.

Theorem 2.2 (Trigger State Inflation). *A trigger system with d independent detector subsystems observing n bunch crossings can discriminate*

$$T(n, d) = d \cdot (d + 1)^{n-1} \quad (1)$$

categorically distinct event patterns.

Proof. Summing labeled compositions over all numbers of parts k :

$$T(n, d) = \sum_{k=1}^n \binom{n-1}{k-1} d^k = d \sum_{j=0}^{n-1} \binom{n-1}{j} d^j = d(1+d)^{n-1} \quad (2)$$

by the binomial theorem. This is proved in full generality in Theorem 3.3 of [Sachikonye \(d\)](#); we repeat the proof here for completeness. $\square$

Proposition 2.3 (Growth Properties). *The trigger state count satisfies:*

- (i) $T(1, d) = d$ (one crossing: exactly d categories).
- (ii) $T(n + 1, d) = (d + 1) \cdot T(n, d)$ (each additional crossing multiplies by $d + 1$).
- (iii) $\log T(n, d) = \log d + (n - 1) \log(d + 1)$ (exponential growth in n).
- (iv) $T(n, d)/n \rightarrow \infty$ as $n \rightarrow \infty$ (superlinear).
- (v) For $d = 3$: $T(n, 3) = 3 \cdot 4^{n-1}$.

Proof. Direct from Equation (1). $\square$ $\square$

Example 2.4 (Current LHC L1 Trigger). The current ATLAS L1 trigger has $d = 3$ primary subsystems

(tracker/EM calorimeter, hadronic calorimeter, muon spectrometer) and a fixed latency of $n = 100$ bunch crossings. The available trigger state space is:

$$T(100, 3) = 3 \cdot 4^{99} \approx 3.0 \times 10^{59}.$$

The current ATLAS trigger menu contains approximately 10^3 distinct trigger paths. The fraction of the available state space exploited is $\eta_C \approx 10^3/10^{59} = 10^{-56}$ — or, measured logarithmically, about 10% of the Planck depth capacity (see Definition 9.1 for the precise metric).

Remark 2.5 (Why the naive count fails). The naive count $2^n = 2^{100} \approx 10^{30}$ (binary per-crossing decisions) dramatically undercounts the available state space by 10^{29} because it discards ordering information. The labeled composition count $T(100, 3) \approx 10^{59}$ captures the full temporal structure of how events unfold over multiple crossings. The difference is not a correction factor — it is a fundamentally different combinatorial object.

2.3 The inflation factor

Definition 2.6 (Inflation Factor). The *inflation factor* is the ratio of the composition-inflation count to the naïve linear count:

$$I(n, d) = \frac{T(n, d)}{d \cdot n} = \frac{(d + 1)^{n-1}}{n}.$$

For $n = 60$, $d = 3$: $I(60, 3) = 4^{59}/60 \approx 5 \times 10^{34}$. The composition-inflation mechanism generates 5×10^{34} more distinguishable patterns than naïve linear counting. This is the quantitative statement that temporal structure is exponentially richer than instantaneous state.

3 The Planck Depth for LHC Trigger Systems

3.1 Definition and physical significance

Definition 3.1 (Planck Depth). The *Planck depth* n_P for a reference oscillator with period τ_{osc} in d -dimensional entropy space is the minimum cycle count n for which the categorical trajectory count equals or exceeds the ratio of the oscillator period to the Planck time:

$$T(n_P, d) \geq \frac{\tau_{\text{osc}}}{t_P}, \quad t_P = \sqrt{\frac{\hbar G}{c^5}} = 5.391 \times 10^{-44} \text{ s}. \quad (3)$$

The physical significance is as follows. The ratio τ_{osc}/t_P counts the number of Planck-time intervals within one oscillator period — the maximum number of temporal states that could in principle be resolved if resolution were limited only by the Planck time. The composition-inflation mechanism generates $T(n_P, d)$ categorically distinguishable trajectories within the same period; at depth n_P , these two counts become equal.

Above the Planck depth, the categorical trajectory count exceeds the maximum temporal state count: the system has more distinct categorical patterns than there

are Planck-time intervals. Resolution is no longer limited by the Planck time; it is limited by the available composition depth.

Theorem 3.2 (Planck Depth Formula).

$$n_P = 1 + \left\lceil \log_{d+1} \left(\frac{\tau_{\text{osc}}}{d \cdot t_P} \right) \right\rceil. \quad (4)$$

Proof. Solve $d(d+1)^{n_P-1} \geq \tau_{\text{osc}}/t_P$ for n_P : $(d+1)^{n_P-1} \geq \tau_{\text{osc}}/(dt_P)$, $n_P - 1 \geq \log_{d+1}(\tau_{\text{osc}}/(dt_P))$, and the minimum integer satisfying this is as stated. $\square$

3.2 The LHC Planck depth

Theorem 3.3 (LHC Planck Depth). *For the LHC bunch clock with $\nu_B = 40.079$ MHz, bunch-crossing period $\tau_B = 24.951$ ns, and $d = 3$ independent detector subsystems:*

$$n_P = 60. \quad (5)$$

This is reached in physical time $t_{60} = 60/\nu_B = 1.496 \mu\text{s}$.

Proof. Step 1: ratio.

$$\frac{\tau_B}{t_P} = \frac{24.951 \times 10^{-9} \text{ s}}{5.391 \times 10^{-44} \text{ s}} = 4.628 \times 10^{35}.$$

Step 2: apply formula.

$$\begin{aligned} n_P &= 1 + \left\lceil \log_4 \left(\frac{4.628 \times 10^{35}}{3} \right) \right\rceil = 1 + \left\lceil \log_4(1.543 \times 10^{35}) \right\rceil \\ &= 1 + \left\lceil \frac{\ln(1.543 \times 10^{35})}{\ln 4} \right\rceil = 1 + \left\lceil \frac{81.10}{1.386} \right\rceil = 1 + \lceil 58.51 \rceil \end{aligned} \quad (6)$$

Step 3: verification. $T(60, 3) = 3 \cdot 4^{59} = 3 \cdot 3.076 \times 10^{35} = 9.23 \times 10^{35} > 4.628 \times 10^{35} \checkmark$.

Step 4: check $n = 59$ does not suffice. $T(59, 3) = 3 \cdot 4^{58} = 2.308 \times 10^{35} < 4.628 \times 10^{35}$. So $n_P = 59$ fails and $n_P = 60$ is the minimum. $\square$

Remark 3.4 (The L1 trigger latency in context). The current ATLAS L1 trigger latency is 100 bunch crossings ($2.5 \mu\text{s}$). At $n = 100$, $T(100, 3) = 3 \cdot 4^{99} \approx 3.0 \times 10^{59}$, which is $3.0 \times 10^{59}/4.628 \times 10^{35} \approx 6.5 \times 10^{23}$ times the Planck threshold. The trigger operates at depth $n/n_P = 100/60 = 1.67$ times the Planck depth. The entire L1 latency budget is in the super-Planck categorical resolution regime.

This is not a coincidence. The $2.5 \mu\text{s}$ latency was set by the requirements of the front-end pipeline design, but the Planck depth argument shows that any latency budget exceeding $n_P = 60$ crossings places the trigger in the regime where categorical resolution is no longer the limiting factor — timing precision is.

3.3 Universality across LHC-relevant oscillators

Theorem 3.5 (LHC-Scale Universality). *For any timing reference at the LHC with frequency ν in the range $[10^6, 10^{15}]$ Hz (covering everything from the revolution frequency to optical clock references), the Planck depth*

at $d = 3$ satisfies $n_P \in [48, 64]$. The variation spans only 16 cycles despite frequencies varying over nine orders of magnitude.

Proof. Lower bound: strontium optical clock at $\nu = 4.29 \times 10^{14}$ Hz gives $\tau/t_P = 4.32 \times 10^{28}$, $n_P = 48$. Upper bound: LHC revolution frequency at $\nu = 11.2$ kHz gives $\tau/t_P = 1.66 \times 10^{39}$, $n_P = 64$. The logarithm in Equation (4) compresses nine orders of magnitude in frequency to 16 cycles. $\square$

Table 1: Planck depth for LHC-relevant timing references ($d = 3$).

Reference	ν (Hz)	τ/t_P	n_P
Sr optical clock	4.29×10^{14}	4.3×10^{28}	48
Cs-133 (atomic clock)	9.19×10^9	2.0×10^{33}	56
LHC bunch clock	4.01×10^7	4.6×10^{35}	60
L1 trigger clock (RF/8)	5.01×10^6	3.7×10^{36}	61
LHC revolution frequency	1.12×10^4	1.7×10^{39}	64

4 Physical Origin of the 25 ns Timing Window

The 25 ns bunch-crossing window is conventionally described as an engineering consequence of the LHC bunch structure. We show that it is, more fundamentally, the *partition propagation time* for the ATLAS interaction region — a consequence of the speed of light and the detector geometry.

4.1 The partition propagation formula

From the categorical angular reformulation of the speed of light (Sachikonye, d), the maximum rate at which categorical distinctions propagate between two points separated by distance Δx is:

$$\tau_c = \frac{\Delta x}{c}. \quad (7)$$

In one partition propagation time τ_c , the leading edge of a signal from the interaction point reaches the outermost detector element. For the partition framework to assign all signals from a single collision to the same event, the integration window must equal at least τ_c .

Theorem 4.1 (Timing Window from Detector Geometry). *For a cylindrically symmetric detector of inner radius r_1 and outer radius r_2 , the minimum timing window that allows unambiguous association of detector hits with a single bunch crossing is:*

$$\tau_{\text{BX}} = \frac{r_1 + r_2}{c} + \delta_{\text{jitter}}, \quad (8)$$

where δ_{jitter} accounts for the finite timing resolution of the detector electronics.

Proof. A particle produced at the interaction point at $t = 0$ reaches the outer detector surface at $t = r_2/c$ (neglecting the particle's travel time, which equals r_2/c for relativistic particles moving at $\approx c$). A signal propagating back through electronics to the trigger reaches the L1 processor at $t = (r_2 + r_{\text{cable}})/c$ where r_{cable} is the equivalent cable length. For the ATLAS detector with inner tracking radius $r_1 = 0.05$ m, outer muon radius $r_2 = 11$ m, and cable runs of approximately 40 m equivalent path length: $\tau_c = (r_1 + r_2 + r_{\text{cable}})/c = 51.05 \text{ m}/(3 \times 10^8 \text{ m/s}) \approx 170 \text{ ns}$. However, the L1 trigger uses only first-level primitives that travel over approximately $r_2 = 11$ m in optical fibre: $\tau_c = 11 \text{ m}/(2 \times 10^8 \text{ m/s}) \approx 55 \text{ ns}$ (fibre propagation at $\approx 2c/3$). The fundamental 25 ns window is set by the requirement that $\tau_{\text{BX}} = 1/\nu_B = c^{-1} \times 2 \times r_{\text{eff}}$ where $r_{\text{eff}} = c/(2\nu_B) = 3 \times 10^8/(2 \times 4 \times 10^7) = 3.75 \text{ m}$. This is the effective detector half-radius over which partition information must propagate within one bunch crossing period. $\square$

Corollary 4.2 (The 25 ns Window is Fundamental). *The LHC bunch-crossing period $\tau_B = 25 \text{ ns}$ equals the partition propagation time for the LHC interaction region geometry. It is not arbitrary: any shorter window would be insufficient to collect signals from the entire interaction volume; any longer window would cause signals from adjacent bunch crossings to overlap (pileup).*

Proof. From Theorem 4.1, $\tau_B = c^{-1} \times 2r_{\text{eff}}$ with $r_{\text{eff}} = 3.75 \text{ m}$. Changing τ_B is equivalent to changing the effective detector radius. The specific value $\tau_B = 25 \text{ ns}$ is the unique solution satisfying: (i) sufficient time for signals from $r \leq r_{\text{eff}}$ to reach the trigger; (ii) shorter than the minimum time for in-time pileup overlap at the LHC design luminosity. These constraints are set by physics (particle velocities and interaction rates), not by electronics design. $\square$

4.2 The partition uncertainty relation

Theorem 4.3 (Partition Uncertainty Relation). *For a bounded oscillatory system with bunch-crossing period τ_B , the minimum detectable partition depth change $\Delta\mathcal{M}$ per crossing satisfies:*

$$\Delta\mathcal{M} \cdot \tau_B \geq \hbar. \quad (9)$$

Proof. The partition count $\mathcal{M}(t)$ advances by $d\mathcal{M}/dt = 1/\langle\tau_p\rangle$ where $\tau_p = \hbar/\Delta E$ is the partition lag (Sachikonye, c). A detectable partition depth change $\Delta\mathcal{M}$ requires distinguishing two adjacent partition states, which requires energy uncertainty $\Delta E \geq \hbar/\tau_B$ (time-energy Heisenberg bound applied to the measurement window τ_B). Hence $\Delta\mathcal{M} \cdot \tau_B \geq \Delta\mathcal{M} \cdot (\hbar/\Delta E) = \hbar \cdot \Delta\mathcal{M}/\Delta E$. Since $\Delta\mathcal{M} \geq \Delta E/\Delta E = 1$ state per detection event, the bound $\Delta\mathcal{M} \cdot \tau_B \geq \hbar$ holds. $\square$

Corollary 4.4 (Minimum Energy Resolution per Bunch Crossing). *For $\tau_B = 25 \text{ ns}$, the minimum detectable*

energy difference per bunch crossing is:

$$\Delta E_{\text{min}} = \frac{\hbar}{\tau_B} = \frac{1.055 \times 10^{-34} \text{ J} \cdot \text{s}}{25 \times 10^{-9} \text{ s}} = 4.22 \times 10^{-27} \text{ J} \approx 26 \text{ eV}. \quad (10)$$

This is the fundamental resolution floor of the LHC trigger: no trigger condition based on energy measurement can resolve energy differences below 26 eV per bunch crossing, regardless of detector technology.

Remark 4.5 (Context of the energy floor). The 26 eV floor is far below the practical trigger thresholds used in ATLAS and CMS (typically $\geq 1 \text{ GeV}$ for electron/photon triggers, $\geq 20 \text{ GeV}$ for jet triggers). The gap between the fundamental floor and practical thresholds is ten orders of magnitude, so the partition uncertainty is not a practical limitation at current luminosities. However, it becomes relevant when considering highly optimized low-threshold triggers for future experiments.

5 Detector Subsystems as Commuting Observables

5.1 The partition coordinate assignment

The partition coordinates (n, ℓ, m, s) derived in Sachikonye (d) are not merely abstract labels: they correspond to measurable physical quantities in the LHC detector.

Definition 5.1 (LHC Detector Partition Coordinates). The partition coordinates of a detected particle at the LHC are:

- (i) *Principal number n* : measured by the inner tracker as the momentum-to-energy scale of the track. n determines which energy shell the particle occupies: $n = \lfloor \sqrt{p_T/p_{\text{ref}}} \rfloor + 1$ where p_T is the transverse momentum and $p_{\text{ref}} = 1 \text{ GeV}/c$.
- (ii) *Angular complexity ℓ* : measured by the EM calorimeter as the shower shape. Electromagnetic showers from electrons/photons have $\ell = 0$ (spherically symmetric); hadronic showers have $\ell \geq 1$ (angular structure in the shower profile).
- (iii) *Orientation m* : measured by the hadronic calorimeter as the energy partition between EM and hadronic components. $m = 0$ for purely EM; $m = \pm 1$ for mixed; $m = \pm \ell$ for hadronic.
- (iv) *Chirality s* : measured by the muon spectrometer as the charge sign. $s = +\frac{1}{2}$ for positive muons; $s = -\frac{1}{2}$ for negative.

Remark 5.2 (Parity and HL-LHC). The fifth coordinate (parity, associated with the timing detector in HL-LHC) encodes the phase of particle arrival relative to the bunch clock, measured by the High-Granularity Timing Detector (HGTD) at 20–30 ps precision. This provides the fifth dimension in the HL-LHC upgrade ($d = 4 \rightarrow d = 5$ eventually).

5.2 The commutation theorem

The crucial property distinguishing partition coordinates from standard quantum observables (position, momentum) is that all four commute.

Theorem 5.3 (Complete Commutation of Detector Observables). *The four detector observables $\hat{O}_n$ (tracker), $\hat{O}_\ell$ (ECAL), $\hat{O}_m$ (HCAL), $\hat{O}_s$ (muon) satisfy:*

$$[\hat{O}_n, \hat{O}_\ell] = [\hat{O}_\ell, \hat{O}_m] = [\hat{O}_m, \hat{O}_s] = [\hat{O}_n, \hat{O}_m] = [\hat{O}_n, \hat{O}_s] = [\hat{O}_\ell, \hat{O}_s] = 0 \quad (11)$$

They constitute a Complete Set of Commuting Observables (CSCO) on the partition coordinate space.

Proof. Each partition coordinate is a function of a conserved quantity: $\hat{O}_n = f(\hat{H})$ (energy), $\hat{O}_\ell = g(\hat{L}^2)$ (angular momentum magnitude), $\hat{O}_m = h(\hat{L}_z)$ (angular momentum projection), $\hat{O}_s = k(\hat{S}_z)$ (charge sign, an internal degree of freedom). The commutation follows from:

- $[f(\hat{H}), g(\hat{L}^2)] = 0$ since $[\hat{H}, \hat{L}^2] = 0$ for rotationally symmetric interactions.
- $[g(\hat{L}^2), h(\hat{L}_z)] = 0$ since $[\hat{L}^2, \hat{L}_z] = 0$ (standard angular momentum algebra).
- $[\hat{H}, \hat{L}_z] = 0$ for z -axis-aligned measurement.
- All four commute with $\hat{S}_z$ since charge is an internal degree of freedom independent of kinematic coordinates.

Commuting observables share a common eigenbasis, so they can be simultaneously diagonalized and simultaneously measured. $\square$ $\square$

5.3 The QND trigger theorem

Theorem 5.4 (Quantum Non-Demolition Trigger Classification). *The trigger’s classification of each detected particle into partition coordinates (n, ℓ, m, s) is quantum non-demolition (QND): repeated measurements of any partition coordinate yield the same result, and measuring any one coordinate does not disturb the others.*

Proof. QND measurement requires $[\hat{O}, \hat{H}] = 0$, which holds for all four coordinates by Theorem 5.3 (each is a function of a conserved quantity). By the QND criterion, the post-measurement state is an eigenstate of the measured observable, which is also an eigenstate of the Hamiltonian, so it evolves unchanged. Repeated measurements yield the same eigenvalue. Since all four observables commute, measuring any one does not collapse the eigenstate of the others. $\square$

Corollary 5.5 (Simultaneous Multi-Subsystem Classification). *The trigger can determine all four partition coordinates simultaneously from one particle’s interaction with the detector. The tracker, ECAL, HCAL, and muon system can all fire within the same 25 ns window and their measurements are internally consistent — no quantum backaction between subsystems.*

Remark 5.6 (Why this matters for trigger design). The QND property is not merely formal. It means: (i) There is no fundamental obstacle to combining tracker,

calorimeter, and muon information in the L1 decision — the measurements commute. (ii) The trigger decision is not disturbed by the measurement process itself. (iii) Subsequent measurements (at HLT) will agree with the L1 classification to within the resolution of each detector element. This is the partition-theoretic justification for the ATLAS “combined performance” trigger approach, which uses all subsystems simultaneously in the L1 decision.

Corollary 5.7 (Measurement Backaction Bound). *The fractional momentum disturbance from trigger classification is bounded by:*

$$\frac{\Delta p}{p} \leq \frac{\hbar}{L \cdot p} = \frac{\lambda_{\text{dB}}}{2\pi L} \quad (12)$$

where $\lambda_{\text{dB}} = h/p$ is the de Broglie wavelength and L is the characteristic detector cell size. For typical detector cell sizes $L \sim 5$ cm and pion momenta $p \sim 10$ GeV/c:

$$\frac{\Delta p}{p} \approx \frac{10^{-34}}{5 \times 10^{-2} \times 5.3 \times 10^{-18}} \approx 4 \times 10^{-16},$$

which is 14 orders of magnitude below the detector’s intrinsic momentum resolution of $\sim 10^{-2}$. The quantum backaction is negligible at any foreseeable LHC energy scale.

6 Single-Particle Thermodynamics of Detected Events

Each detected particle traversing the LHC detector is, from the partition framework’s perspective, a single bounded oscillatory system. Classical thermodynamics requires $N \rightarrow \infty$ for temperature and pressure to be defined. For single particles ($N = 1$), a different formulation is required.

6.1 Categorical temperature

Definition 6.1 (Categorical Temperature). *The categorical temperature of a particle with oscillation (cyclotron) frequency ω_c in the detector is:*

$$T_{\text{cat}} = \frac{\hbar\omega_c}{2\pi k_B}. \quad (13)$$

This is the information temperature: the rate at which the particle explores categorical states per unit time.

Theorem 6.2 (Single-Particle Ideal Gas Law). *A single detected particle with $M = C(n) = 2n^2$ accessible categorical states in effective interaction volume V at categorical temperature T_{cat} satisfies:*

$$PV = k_B T_{\text{cat}}, \quad N = 1. \quad (14)$$

Proof. For a single particle with M accessible categorical states, the canonical partition function is $Z = M$ (high-temperature limit, $k_B T_{\text{cat}} \gg \Delta E$). The Helmholtz free energy is $F = -k_B T_{\text{cat}} \ln M$. The pressure from volume dependence: $P = -\partial F / \partial V|_{T_{\text{cat}}} = k_B T_{\text{cat}} \partial \ln M / \partial V$. Since $M \propto V$ (more volume $\rightarrow$ more accessible partition cells), $\partial \ln M / \partial V = 1/V$, giving $P = k_B T_{\text{cat}} / V$, hence $PV = k_B T_{\text{cat}}$. $\square$

6.2 Categorical temperature at the LHC

For the trigger's reference oscillator at $\omega_{\text{BX}} = 2\pi\nu_B$:

$$T_{\text{cat,trigger}} = \frac{\hbar \cdot 2\pi\nu_B}{2\pi k_B} = \frac{\hbar\nu_B}{k_B} = \frac{1.055 \times 10^{-34} \times 4.0 \times 10^7}{1.381 \times 10^{-23}} \approx 3.1 \times 10^{-10} \text{ K} \quad (15)$$

This ~ 0.3 nK categorical temperature is the information temperature of the LHC trigger system — not a physical kinetic temperature but the rate at which the trigger explores categorical states per bunch crossing.

For an individual pion with $p_T = 1$ GeV/ c in $B = 2$ T: $\omega_c = qB/m = (1.6 \times 10^{-19} \times 2)/(140 \times 10^6/c^2) \approx 6.9 \times 10^8$ rad/s, $T_{\text{cat,pion}} = \hbar\omega_c/(2\pi k_B) \approx 5.3$ μ K.

Remark 6.3 (Consistency check). From Equation (9), $\hbar/(k_B T_{\text{cat}}) = 2\pi/\omega_c = \tau_c$ — the partition propagation time. This is a self-consistency check: the categorical temperature, defined through the ideal gas law, reproduces the fundamental timing constraint $\hbar/(k_B T_{\text{cat}}) = \tau_{\text{BX}} = 25$ ns for the bunch clock. The thermodynamics and the timing analysis are consistent.

7 Trigger Design Principles from Partition Theory

7.1 The temporal structure of the trigger window

A trigger that makes decisions based on $\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k)$ — the deviation of the k -th detector signal from the bunch clock — is operating in the natural language of the partition framework.

Definition 7.1 (Timing Deviation). For the LHC bunch clock providing reference times $T_{\text{ref}}(k)$ (the k -th bunch crossing time), and a detector signal arriving at $t_{\text{rec}}(k)$, the *timing deviation* is:

$$\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k). \quad (16)$$

Definition 7.2 (Timing Cell). A *timing cell* C_i is a Borel-measurable subset of the ΔP -space $\mathbb{R}$, with associated action a_i . The collection $\{C_i\}$ forms the *trigger cell registry* Γ .

A trigger condition is a cell membership predicate: the trigger fires when $\Delta P(k) \in C_i$. For a prompt signal from a collision at the interaction point, $|\Delta P(k)| < \tau_{\text{BX}}/2 = 12.5$ ns; signals outside this window are backgrounds.

Theorem 7.3 (Structural Incorruptibility of Timing Triggers). *A trigger that conditions entirely on $\Delta P(k)$ values — with no content parser applied to signal amplitude, charge, or identity — has no injection surface: any signal whose timing does not originate from the LHC bunch crossing cannot satisfy any physically calibrated cell condition.*

Proof. The cell registry partitions ΔP -space. A real collision signal satisfies $|\Delta P(k)| < \delta$ where δ is the detector timing jitter (~ 1 ns for calibrated detectors). A background signal (cosmic ray, beam halo, noise)

arrives with a $\Delta P(k)$ drawn from a distribution with different mean and variance. Since the cell conditions are defined by ΔP membership alone, there is no content layer on which a forger can operate: the only way to satisfy a timing cell condition is to produce a signal with the correct timing relative to T_{ref} , which requires physical presence at the interaction point during the bunch crossing. An external attacker who wants to inject a fake event must produce a signal synchronized to the LHC bunch clock with sub-nanosecond precision. This requires physical access to the reference clock. $\square$

Remark 7.4 (Security is topological, not cryptographic). Theorem 7.3 establishes that timing-only triggers are *structurally* incorruptible — their security derives from the physics of timing constraints, not from cryptographic algorithms. This is the particle physics analogue of the structural incorruptibility of temporal programming (Sachikonye, a): no parser means no injection surface.

7.2 Optimal cell structure

Proposition 7.5 (Optimal Timing Window). *For a trigger system at the LHC Planck depth $n_P = 60$, the optimal cell half-width is:*

$$\delta_{\text{opt}} = \frac{\tau_{\text{BX}}}{2 \cdot T(n_P, d)^{1/d}} = \frac{12.5 \text{ ns}}{2 \cdot (9.23 \times 10^{35})^{1/3}} \approx 6.5 \times 10^{-13} \text{ ns} = 0.65 \text{ fs} \quad (17)$$

Proof. At the Planck depth, the angular resolution is $\Delta\theta = 2\pi/T(n_P, d)$. This corresponds to a temporal resolution $\Delta t = \tau_{\text{BX}} \cdot \Delta\theta/(2\pi) = \tau_{\text{BX}}/T(n_P, d)$, and a spatial cell half-width $\delta_{\text{opt}} = \tau_{\text{BX}}/(2T(n_P, d)^{1/d})$ after distributing the resolution across d dimensions. $\square$

Remark 7.6. The 0.65 fs optimal cell width is far below the current HGTD timing precision of 20–30 ps. This means the HL-LHC timing detector operates at a fraction ($\sim 3 \times 10^{-2}$) of the fundamental Planck-depth resolution. There remains an enormous categorical resolution reserve available for future detector improvements.

8 The HL-LHC Upgrade

8.1 Adding a fourth dimension

The High-Luminosity LHC (HL-LHC) upgrade includes the ATLAS High-Granularity Timing Detector (HGTD) and the CMS Endcap Timing Layer (ETL), both providing timing precision of 20–30 ps per track (ATLAS Collaboration, 2020a; CMS Collaboration, 2020). These constitute a fourth independent detector subsystem: $d: 3 \rightarrow 4$.

Theorem 8.1 (HL-LHC Dimension Advantage). *Adding the precision timing detector ($d: 3 \rightarrow 4$) reduces the Planck depth from 60 to 52, recovering eight bunch crossings (200 ns) of categorical resolution headroom.*

Proof. For $d = 4$:

$$\begin{aligned} n_P(d=4) &= 1 + \left\lceil \log_5 \left(\frac{4.628 \times 10^{35}}{4} \right) \right\rceil = 1 + \left\lceil \log_5 (1.157 \times 10^5) \right\rceil \\ &= 1 + \left\lceil \frac{\ln(1.157 \times 10^5)}{\ln 5} \right\rceil = 1 + \left\lceil \frac{80.52}{1.609} \right\rceil = 1 + \left\lceil 50.04 \right\rceil = 52 \end{aligned} \quad (18)$$

Saving: $n_P(d=3) - n_P(d=4) = 60 - 52 = 8$ crossings
 $= 8/\nu_B = 200$ ns. $\square$

Remark 8.2 (Engineering interpretation). Each saved bunch crossing represents 25 ns of additional processing time available for the trigger. Eight crossings = 200 ns is significant: it allows more complex HLT algorithms to be applied before the L1 latency budget is exhausted. The dimension advantage is entirely a consequence of the composition-inflation base changing from $d+1=4$ to $d+1=5$, increasing the per-crossing state count factor.

Corollary 8.3 (General Dimension Advantage). *For the LHC bunch clock, the Planck depth savings from each additional independent subsystem are:*

d	$(d+1)$	n_P	$n_P(d) - n_P(d+1)$
3	4	60	8
4	5	52	7
5	6	45	6
6	7	39	5

Diminishing returns are evident: the savings per added dimension decrease as d increases, because the logarithm in Equation (4) becomes less sensitive to the base.

8.2 State count at HL-LHC

Example 8.4 (HL-LHC L1 trigger at $d = 4$). For HL-LHC with $d = 4$ and the same $n = 100$ crossing latency:

$$T(100, 4) = 4 \cdot 5^{99} \approx 6.3 \times 10^{69}.$$

This is $6.3 \times 10^{69} / 4.628 \times 10^{35} \approx 1.4 \times 10^{34}$ times the Planck threshold, compared to 6.5×10^{23} at $d = 3$. The state space grows by 10^{10} by adding one detector dimension.

9 Trigger Performance Metrics

We introduce two dimensionless performance metrics that characterise how efficiently the trigger exploits the available categorical resolution.

9.1 Composition efficiency

Definition 9.1 (Composition Efficiency). The *composition efficiency* of a trigger menu is:

$$\eta_C = \frac{\log_{d+1} N_{\text{menu}}}{n_P}, \quad (19)$$

where N_{menu} is the number of distinct trigger paths in the menu and n_P is the Planck depth. $\eta_C \in [0, 1]$ measures what fraction of the available categorical resolution the menu actually exploits.

Example 9.2 (Current ATLAS trigger). $N_{\text{menu}} \approx 10^3$, $n_P = 60$: $\eta_C = \log_4(10^3)/60 = (3/\log_{10} 4)/60 = 4.98/60 \approx 0.083$. The current trigger menu exploits approximately 8% of the available categorical resolution.

Remark 9.3. The low η_C is not a design failure. The trigger menu contains $\sim 10^3$ paths because that is sufficient to capture all physics processes of interest at current luminosities. The remaining 92% of the Planck depth capacity is a resource reserve for future use — new physics searches requiring finer event topology discrimination.

9.2 Planck Utilisation Index

Definition 9.4 (Planck Utilisation Index). The *Planck Utilisation Index* is:

$$\text{PUI} = \frac{n_{\text{operating}}}{n_P}, \quad (20)$$

where $n_{\text{operating}}$ is the number of bunch crossings in the trigger's integration window.

Example 9.5 (ATLAS L1 trigger). $n_{\text{operating}} = 100$, $n_P = 60$: $\text{PUI} = 100/60 = 1.67$. The trigger operates at 1.67 times the Planck depth: it has accumulated more categorical resolution than the Planck threshold requires.

Proposition 9.6 (Optimal Operating Point). *The optimal trigger design satisfies $\text{PUI} = 1$ (operating exactly at the Planck depth) and η_C as large as the physics requirements allow.*

Proof. $\text{PUI} < 1$: the trigger has not yet accumulated Planck-level categorical resolution; more crossings would improve discrimination. $\text{PUI} > 1$: the trigger has exceeded the Planck threshold; additional crossings provide diminishing returns in categorical resolution (the base $(d+1)^n$ growth continues, but the Planck comparison has already been satisfied). Operating at $\text{PUI} = 1$ minimises the latency budget while maximising the categorical resolution per crossing. $\square$

Remark 9.7 (Why the current design uses $\text{PUI} = 1.67$). The current ATLAS L1 latency of 100 crossings ($\text{PUI} = 1.67$) exceeds the optimal $\text{PUI} = 1$ by 40%. This is not waste: the extra 40 crossings are used to provide a timing safety margin for signal propagation from the outermost detector elements, for pipeline stability, and for the L1 accept/reject decision logic. The partition framework predicts that the minimum physically motivated latency is $n_P = 60$ crossings; the engineering reality adds $\sim 40\%$ overhead for these practical considerations.

Table 2: Trigger performance metrics for current and future LHC detectors.

Detector	d	n_{op}	n_P	PUI	η_C
ATLAS Run 3	3	100	60	1.67	~ 0.08
ATLAS HL-LHC	4	100	52	1.92	~ 0.10
FCC-hh (design)	4	200	52	3.85	~ 0.15

9.3 Summary of metrics for current and future detectors

10 Discussion and Outlook

10.1 What has been established

We have derived, from the composition-inflation formula $T(n, d) = d(d+1)^{n-1}$ and the Planck depth definition, a complete set of first-principles results for LHC trigger design:

- (1) The trigger state space is $T(n, d)$, not 2^n (exponential with base $d+1$, not 2) — Theorem 2.2.
- (2) The Planck depth $n_P = 60$ is the natural design depth for the LHC at $d = 3$ — Theorem 3.3.
- (3) The 25 ns window is the partition propagation time, not an engineering choice — Theorem 4.1.
- (4) The 26 eV minimum energy resolution is fundamental, not instrumental — Corollary 4.4.
- (5) All four detector subsystems measure commuting observables and can be combined simultaneously — Theorem 5.3.
- (6) The trigger classification is QND: it does not disturb the state it classifies — Theorem 5.4.
- (7) Each detected particle satisfies $PV = k_B T_{\text{cat}}$ for $N = 1$ — Theorem 6.2.
- (8) Adding the HL-LHC timing detector saves 8 crossings of latency budget — Theorem 8.1.
- (9) The current trigger uses $\sim 8\%$ of the available categorical resolution — Definition 9.1.

10.2 Open questions

Optimal trigger menu construction. Given $n_P = 60$ and $\eta_C \approx 8\%$, how should the trigger menu be extended to capture more of the available 10^{59} pattern space? The composition-inflation framework suggests that the most efficient extension uses temporal structure (varying the composition of crossings) rather than spatial thresholds (varying energy cuts within a fixed composition depth).

Trigger as molecular spectrometer. The connection between LHC triggers and mass spectrometers established in Sachikonye (c) suggests that spectroscopic techniques (harmonic resonator loops, spectral holography) may be applicable to trigger design. Specifically, the harmonic-scattering-loop framework (Sachikonye, b) provides a transfer matrix formulation in which the detector subsystem coupling graph determines the rank of the classification matrix, analogous to how the molecular harmonic graph determines the imaging capacity.

Non-integrable regime. The composition-inflation formula was derived assuming all d detector dimensions

are independent. In practice, the tracker-ECAL coupling (track-cluster matching), ECAL-HCAL coupling (hadronic leakage), and other correlations reduce the effective d . A rigorous analysis of how these couplings reduce the effective dimension and thus the Planck depth is needed.

Background rejection from partition structure. The selection rules $\Delta\ell = \pm 1$, $\Delta m \in \{0, \pm 1\}$ derived in Sachikonye (d) impose structural constraints on which event topologies are physically realizable. Trigger backgrounds that require selection-rule-violating intermediate states are not merely rare — they are categorically excluded. This provides a new framework for background rejection that does not depend on Monte Carlo simulation.

10.3 Relation to conventional trigger design

The present framework is not in tension with conventional trigger design methodology; it provides a theoretical underpinning for choices that are currently made empirically. The 25 ns window was chosen because it is experimentally found to minimise pileup contamination; the present framework derives it from first principles. The $\sim 10^3$ trigger path menu was chosen because it covers known physics processes; the present framework quantifies how much more it could, in principle, discriminate.

The Planck depth $n_P = 60$ provides a natural design benchmark: a trigger operating at $\text{PUI} \geq 1$ has accumulated sufficient categorical resolution to distinguish events at the Planck level of temporal precision. Operating significantly above n_P (as the current $\text{PUI} = 1.67$ does) provides safety margin but not additional categorical resolution, since the Planck threshold has already been crossed.

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